Exercise 3: Join, union, filtering and aggregates

SQL Joins

1. INNER Join

```
SELECT
    e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
    p.ProjectID, p.ProjectName, p.Budget, p.Status

FROM Employee e
INNER JOIN Project p
ON e.EmployeeID = p.EmployeeID
```

Expected output

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
2	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
4	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2. LEFT Join

```
SELECT
    e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
    p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employee e
LEFT Join Project p
ON e.EmployeeID = p.EmployeeID;
```

Output

EmployeeID	FirstName	LastName	Department	Salary	ProjectId	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

3 Right Join

```
SELECT
    P.ProjectID, P.ProjectName, P.Budget, P.Status,
    e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary
FROM Employee e
RIGHT JOIN Projects P
ON e.EmployeeID = P.EmployeeID;
```

Output

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk management	80000	Pending	6	Michael	Clark	Finance	80000

Full Outer Join

```
SELECT
    e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary
    p.projectID, p.ProjectName, P.Budget, P.Status

FROM Employees e
Full Outer Join Projects p
ON e.EmployeeID = P.EmployeeID;
```

Output

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Research	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

3. Filtering Statements

```
SELECT
    EmployeeID,
    FirstName,
    LastName,
    Department,
    Salary
FROM Employees
WHERE Department IN ('IT', 'Finance')
```

Output

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

4 Filtering Statements

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70000;
```

Output

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

5.

```
SELECT ProjectID, ProjectName, Budget, Status
FROM ~~employees~~ Projects
WHERE Status = 'completed'
```

Output

ProjectID	ProjectName	Budget	Status
102	Employee training	50000	Ongoing
103	Cybersecurity audit	75000	Pending
104	Finance analysis	90000	Ongoing
106	Risk management	80000	Pending

6.

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000
AND Status <> 'Completed'
```

Output

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Advertising Analysis	90000	Ongoing
106	Risk Management	80000	Pending

7.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
       City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;
```

Output

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	8000	New York
3	Bob	Johnson	Marketing	75000	Toronto
1	John	Doe	IT	70000	New York

8.

```
SELECT employeeID, firstname, lastname, department, salary
FROM Employees
ORDER BY salary DESC
LIMIT 3;
```

Output

EmployeeID	Firstname	lastname	department	salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

9.

```
SELECT department, SUM(salary) AS TotalSalary
FROM Employees
Group BY department
Order by TotalSalary DESC;
```

Output

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

```
SELECT City, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65000
```

Output

City	Average Salary
New York	78000
Toronto	75000
London	72000

11.

```
SELECT Department, COUNT(EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) > 1;
```

Output

Department	Employee Count
IT	2
Finance	2

12.

```
SELECT Status, Count(ProjectID) AS ProjectCount
FROM Projects
Group By Status
Having Count(ProjectID) >= 2;
```

Output

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

13.

```
SELECT e.EmployeeID, e.FirstName, e.LastName,
 Sum(p.Budget) AS TotalProjectBudget

FROM Employee e
Join Project p
ON e.EmployeeID = p.EmployeeID
Group BY e.EmployeeID, e.FirstName, e.LastName
Having Sum(p.Budget) > 150000;
```

Output

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000